Expertise-Leveling Technology and the Division of Creative Labor

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Technology and Expertise

A common view of technology

Machines take over some tasks and make specific workers more productive in others.



Personal computers

Technology and Expertise

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Personal computers

Expertise-leveling technologies

Some technologies also let people perform tasks that previously required specialist training.



Photography

Portraits without painting skills

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Calculation without manual methods

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AI

Writing without specialist training

Expertise-leveling technologies: what we don't know

Most evidence examines individual performance.

Studying broader effects is difficult: contributors' tasks are rarely observed, output is hard to quantify, and technology adoption is endogenous.

Potential Impacts of Expertise-Leveling Technologies

Expertise-leveling technologies: what we don't know

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Employment

How many people work?

Expertise-leveling technologies: what we don't know

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Employment

How many people work?



Organization of production

Who contributes, and which tasks do they perform?

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Potential Impacts of Expertise-Leveling Technologies



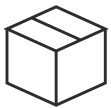
Employment

How many people work?



Organization of production

Who contributes, and which tasks do they perform?



Output

How does the product change?

This paper

How do employment, team organization, and output change when technology lowers the expertise required to perform some tasks?

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Empirical laboratory: the US recorded-music industry

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Musicians can reuse and combine recorded performances without learning to play those parts.

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Technology: The AKAI MPC60 digital sampler, introduced in 1988

Musicians can reuse and combine recorded performances without learning to play those parts.

Identification: The AKAI MPC60 promotional campaign

Demonstrations alongside Michael Jackson's US Bad Tour showed musicians how to use the sampler.

Compare changes before and after the instrument's introduction in exposed locations and locations hosting other major tours.

Preview of the findings

1. Employment

Sampling rises, with **no detectable decline in employment** in music production.

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Artists associated with sampled records are **more likely to be self-taught**.

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More contributors focus exclusively on creative authorship and production, with no detectable broadening of individual task bundles.

3. Expertise

Artists associated with sampled records are **more likely to be self-taught**.

4. Output

Records become **acoustically more differentiated**, consistent with greater innovation in recorded sound, with no detectable change in lyrics.

Contributions to the literature

Technological change and expertise
Autor and Thompson (2025); Althoff and Reichardt (2026); Noy and Zhang (2023); Brynjolfsson et al. (2025)

Causal evidence on how employment, the division of labor, and output jointly adjust over more than a decade as an expertise-leveling technology spreads.

Specialization and the division of labor
Becker and Murphy (1992); Garicano (2000); Dell'Acqua et al. (2026)

Show that lower expertise requirements can increase the number of contributors focused on creative tasks without broadening individual task bundles.

Economics of creative production
Aguiar and Waldfogel (2018); Askin and Mauskapf (2017); Caves (2000)

Show how a production technology changes both the organization of creative labor and product differentiation.

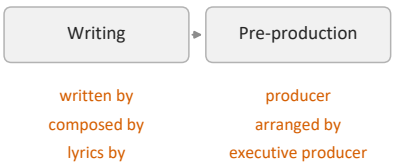
Outline

1. **Setting**
2. **Data and Empirical Strategy**
3. **Results**
   - Diffusion and employment
   - Organization of production
   - Artist backgrounds
   - Output
4. **Conclusion**

Setting

Teams in recorded-music production

Creative authorship and production



▸ Sgt. Pepper's Example

Teams in recorded-music production

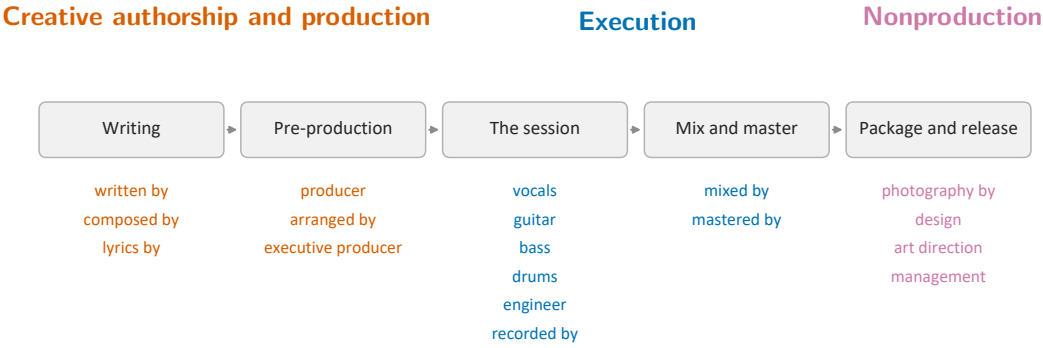
Creative authorship and production | **Execution**



Writing	Pre-production	The session	Mix and master
written by	producer	vocals	mixed by
composed by	arranged by	guitar	mastered by
lyrics by	executive producer	bass	
		drums	
		engineer	
		recorded by	

▸ Sgt. Pepper's Example

Teams in recorded-music production



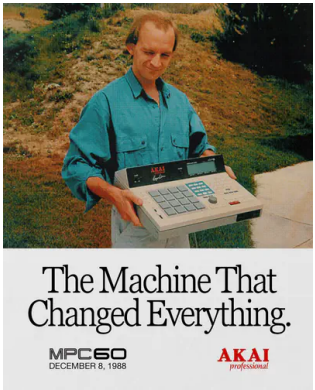
▸ Sgt. Pepper's Example

What is digital sampling?

Three key characteristics

Record

Take a fragment from an existing recording.



The AKAI MPC60 (1988)

▸ Audio example

What is digital sampling?

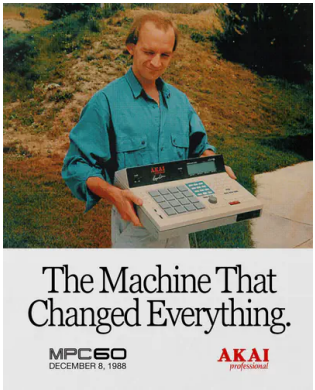
Three key characteristics

Record

Take a fragment from an existing recording.

Transform

Repeat, rearrange, and combine recorded fragments.



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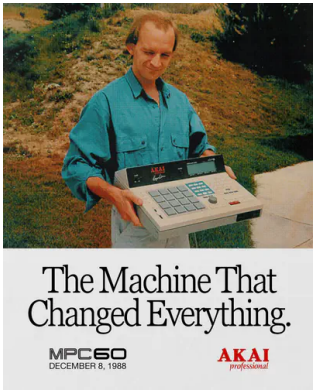
Transform

Repeat, rearrange, and combine recorded fragments.

Build

Use recorded performances as parts of a new recording.

▸ Audio example



The AKAI MPC60 (1988)

The AKAI MPC60, introduced in 1988



The AKAI MPC60 (1988)

Several production tasks in one device

Sample recorded sounds, program rhythms, sequence musical parts, and assemble an arrangement.

The AKAI MPC60, introduced in 1988



The AKAI MPC60 (1988)

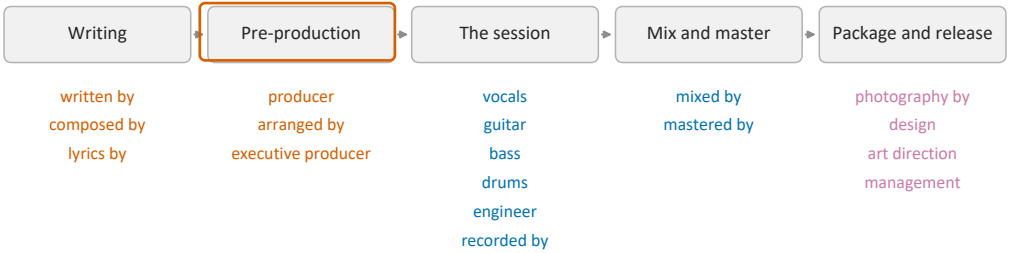
Several production tasks in one device

Sample recorded sounds, program rhythms, sequence musical parts, and assemble an arrangement.

A simple interface

Pressure-sensitive pads and a simple workflow made these functions easier to use.

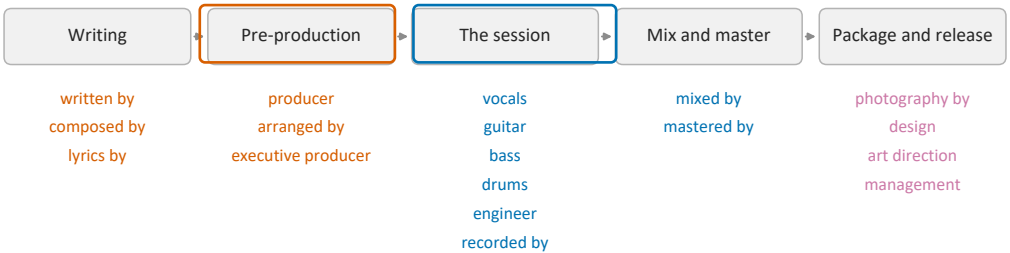
Where sampling acts



Pre-production

The AKAI lets musicians select, sequence, and arrange recorded sounds.

Where sampling acts



Pre-production

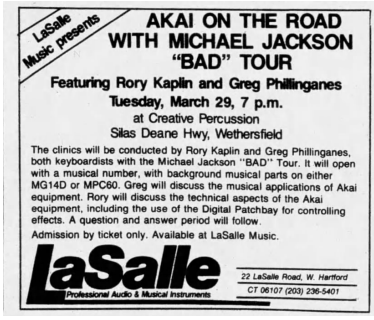
The AKAI lets musicians select, sequence, and arrange recorded sounds.

The session

The AKAI plays back recorded performances, allowing musicians to reuse parts without performing them again.

Michael Jackson's tour carried the sampler campaign





Clinics were hands-on demonstrations where musicians from Jackson's band showed attendees how to use the sampler.

▸ Clinic details ▸ Equipment listings

Data and Empirical Strategy

The universe of records: Discogs

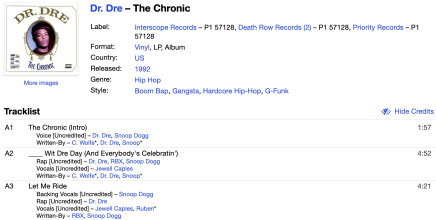


Dr. Dre, *The Chronic* (1992), illustrates the information observed for each record.

70,999 records, 1980–2000

US recording locations across genres and artists

The universe of records: Discogs



Dr. Dre – The Chronic

Label: Interscope Records – P1 57128, Death Row Records (2) – P1 57128, Priority Records – P1 57128
Format: Vinyl, LP, Album
Country: US
Released: 1992
Genre: Hip Hop
Style: Boom Bap, Gangsta, Hardcore Hip-Hop, G-Funk

More images

Tracklist Hide Credits

A1	The Chronic (Intro) Voice [Uncredited] – Dr. Dre, Snoop Dogg Written-By – C. Wolfe*, Dr. Dre, Snoop*	1:57
A2	___ We Dre Day (And Everybody's Celebratin') Rap [Uncredited] – Dr. Dre, RBX, Snoop Dogg Vocals [Uncredited] – Jewell Caples Written-By – C. Wolfe*, Dr. Dre, Snoop*	4:52
A3	Let Me Ride Backing Vocals [Uncredited] – Snoop Dogg Rap [Uncredited] – Dr. Dre Vocals [Uncredited] – Jewell Caples, Ruben* Written-By – RBX, Snoop Dogg	4:21

Companies, etc.

Recorded At – Death Row Studios
Mixed At – Larrabee Sound Studios
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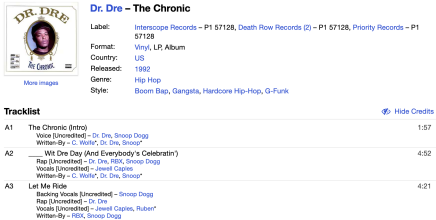
70,999 records, 1980–2000

US recording locations across genres and artists

7,229 US recording studios

Studios locate each record in a local labor market.

The universe of records: Discogs



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- Coverage
- Summary statistics
- Contributor example
- Sampling data
- Additional data

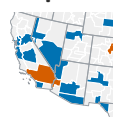
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Example: Los Angeles



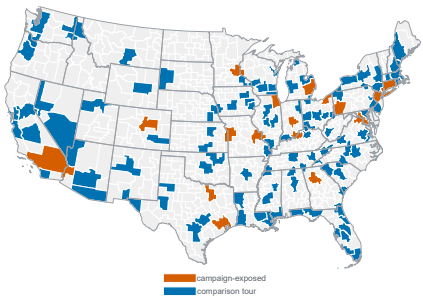
The Chronic: Death Row Studios, Los Angeles.

Sampling, artists' backgrounds, and output

Source	Measure
WhoSampled	Whether a record uses sampling
Wikidata and Wikipedia biographies	Main artists' musical training and education
Spotify Echo Nest and MusicFlamingo	Acoustic differentiation from earlier recordings
Genius	Lyrical differentiation from earlier recordings
Billboard 200	Chart entry, weeks on the chart, and rank-weighted performance

▸ Sampling data ▸ Backgrounds and content ▸ Chart performance

Campaign exposure and comparison locations



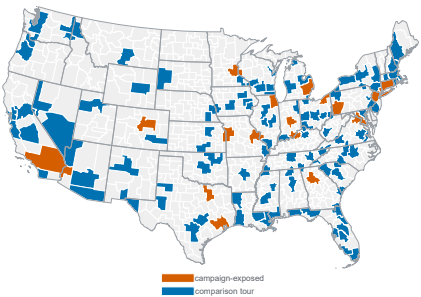
Geographic variation

- **18 exposed CZs** hosted Michael Jackson and the AKAI campaign in 1988.
- **107 comparison CZs** hosted another top-10 tour of 1988.

Time variation

Compare changes before (1980–1988) and after (1989–2000) the campaign.

Campaign exposure and comparison locations



Geographic variation

- **18 exposed CZs** hosted Michael Jackson and the AKAI campaign in 1988.
- **107 comparison CZs** hosted another top-10 tour of 1988.

Time variation

Compare changes before (1980–1988) and after (1989–2000) the campaign.

Identifying assumption: without the campaign, exposed and comparison zones would have followed similar changes in the outcomes.

▸ Studio assignment ▸ Local differences

Empirical strategy

$$Y_i = \sum_{\tau \neq 1988} \beta_\tau \left(\text{Exposed}_{m(i)} \times \mathbf{1}\{t(i) = \tau\} \right) + \alpha_{u(i)} + \rho_{d(i),t(i)} + \varepsilon_i$$

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- i: a record for sampling, organization, and output; a commuting-zone-year for employment. Sample: 1980–2000.

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- Y_i: sampling indicator or local sampling share; contributors per 100,000 residents aged 20–64; contributor counts or task-breadth shares; acoustic or lyrical distance.

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- $\text{Exposed}_{m(i)}$: one for commuting zones hosting Jackson's 1988 tour and campaign. $m(i)$, $d(i)$, and $t(i)$ identify the zone, census division, and year (release year for records).

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- $\alpha_{u(i)}$: studio fixed effects for organization, commuting-zone fixed effects otherwise. $\rho_{d(i),t(i)}$: census-division-by-year fixed effects. ε_i: residual.

β_τ: exposed–comparison difference in year τ, relative to 1988.

Pooled estimates replace the annual interactions with $\text{Exposed}_{m(i)} \times \text{Post}_{t(i)}$, where Post is 1989–2000. CR2 inference, clustered by commuting zone.

What comes next

1. **Diffusion and employment**

Does sampling spread, and does employment decline?

2. **Organization of production**

Do individual roles broaden, or does team composition change?

3. **Artists' backgrounds**

Are artists associated with sampling less formally trained?

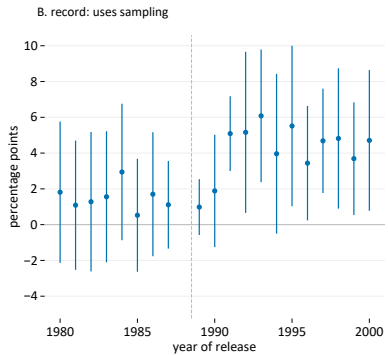
4. **Output**

Does recorded music become more differentiated?

Diffusion and Employment

Does sampling spread, and does employment decline?

Results: Sampling increases

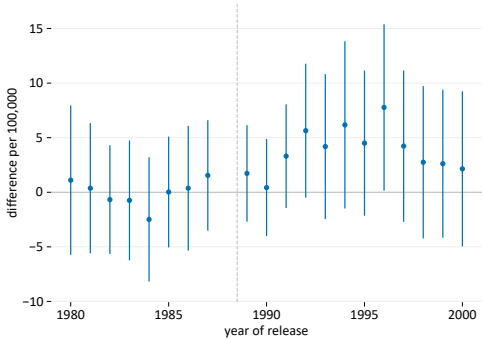


2.98 percentage points: about three additional records using sampling per hundred records.

▸ CZ result ▸ Other tours and local conditions ▸ Diffusion checks ▸ Mix equipment listings

Results: No detectable decline in employment in music production

Distinct contributors appearing in local record-production credits each year, per 100,000 working-age residents.



Pooled change: +3.85 contributors per 100,000 working-age residents ($p = 0.150$).

Employment shows no detectable decline relative to local population.

▸ Contributor categories ▸ Robustness

Organization of Production

Do individual tasks broaden, or does team composition change?

Results: Contributors working only in creative authorship and production

	Contributors per record			
	Creative authorship and production only	Execution only	Nonproduction only	Across two or more task categories
Exposure × *Post*	+0.449**	+0.217	+0.023	−0.132
	(0.160)	(0.196)	(0.071)	(0.145)
Studio FE	Yes	Yes	Yes	Yes
Census division-by-year FE	Yes	Yes	Yes	Yes
Observations	70,994	70,994	70,994	70,994
Mean, 1980–1988	3.37	8.00	1.62	2.25

Note. Pooled difference-in-differences estimates. Post: 1989–2000. CR2 standard errors in parentheses, clustered by commuting zone. Each contributor counts once per record in one of the four groups. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$ (CR2 inference).

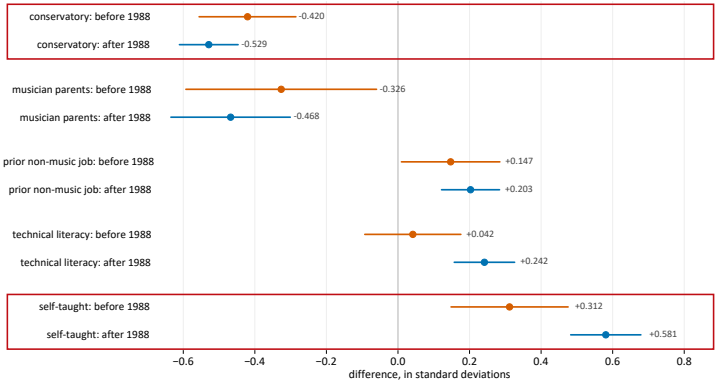
About **13%** more contributors working only in creative authorship and production, with no detectable decline in the other groups.

▸ Event studies ▸ Creative contributions ▸ Team size ▸ Robustness ▸ Sample attribution ▸ Sampling propensity

Artists' Background

Are artists associated with sampling less formally trained?

Results: Artists associated with sampling are less formally trained



After 1988, **65%** of artists associated with sampled records are self-taught, versus **37%** of other artists.

▸ Training and coverage ▸ Full background figures

Output

How does the product change?

Measuring innovation in recorded sound

Following **Askin and Mauskapf (2017)**, I measure a record's acoustic distance from the previous year's releases.

Compare **eleven acoustic characteristics from Spotify Echo Nest**, weighting earlier records by genre proximity (validation using MusicFlamingo).

Higher values indicate **greater acoustic differentiation** from earlier releases.

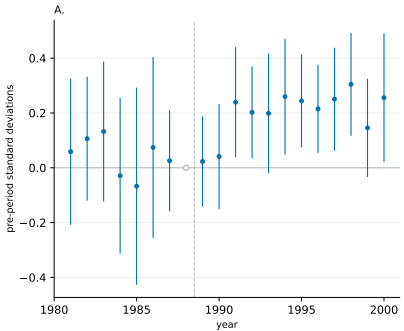
I use this differentiation as a **proxy for innovation in recorded sound**.

I apply the same approach to lyrics, measuring lyrical distance from the previous year's releases.

▸ Distance construction ▸ Measurement checks

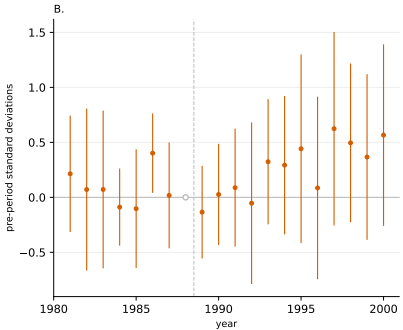
Results: Output becomes more acoustically differentiated

Acoustic distance



Acoustic distance from the previous year's recordings increases by **0.17 standard deviations**.

Lyrical distance



No detectable change in lyrical distance.

▸ Distance measure ▸ Genre comparisons ▸ Content checks ▸ Commercial performance

Robustness checks

Diffusion ▸ Details

Local characteristics, comparison-tour cutoffs, 1988 coding, exposure reassignment, and local economic conditions

Employment ▸ Details

Local characteristics, Poisson counts, population denominators, and entity cleaning

Organization ▸ Samples ▸ Credits and specifications

Comparison locations, release dating, studio assignment, credit classifications and exclusions, timing, genre, format, and reassignment

Output ▸ Details

Studio effects, genre effects, within-genre comparisons, audio representations, lyric length, and coverage

▸ Discussion

Conclusions

Lower expertise requirements can support a greater role for **creative specialists**.

1. **Sampling increases**, with **no detectable decline in employment** in music production.
2. Records gain contributors focused only on **creative authorship and production**, with no detectable broadening of individual task bundles.
3. Artists associated with sampling are **more often self-taught**, consistent with different routes into production.
4. Records become **acoustically more differentiated**, consistent with greater innovation in recorded sound. No detectable change in lyrics.

⇒ **Lower execution requirements can support a greater role for creative specialists and more differentiated output, even without broader individual roles or a detectable employment response.**

Appendix

Discussion

Adjustment in production

Records gained contributors working only in creative authorship and production and became more acoustically differentiated, with no detectable broadening of individual task bundles or employment decline.

Ownership of reused work

Ownership and access terms may affect who benefits when human work becomes an input into later production without its creator participating again.
Cullen, Li, and Li (2026), *Labor as Capital: AI and the Ownership of Expertise*

Complexity of output

Does easier execution allow musicians to realize more elaborate ideas?

The corresponding question for AI

Which tasks become easier, and which contributions do other workers still provide?

Examining employment alone would miss the changes in team composition and output that emerged over the following decade.

Back

One contributor can perform several tasks

Credits

A&R [Director] – John McClain
Art Direction, Design – Kimberly Holt•, Unleashed (2)
Coordinator – Kimberly Brown (2)
Engineer [Assistant Mix] – Willie Will
Executive-Producer, Management – Suge Knight
Flute, Saxophone – Katisse Buckingham
Guitar – Chris Clairmont, Eric (The Drunk) Borders•
Keyboards – Justin Reinhardt
Keyboards, Bass Guitar, Keyboards [Bass] – Colin W
Mastered By – Bernie Grundman
Mixed By – Chris (The Glove) Taylor•
Mixed By, Engineer [Mix] – Greg (Gregski) Royal•
Photography By – Daniel Jordan (4)
Producer, Mixed By, Keyboards – Dr. Dre
Written-By [Uncredited] – Percy Carey

Credits for *The Chronic*, Discogs

Dr. Dre's credits in this panel

Producer	Creative authorship and production
Mixed by	Execution
Keyboards	Execution

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Credits for *The Chronic*, Discogs

Dr. Dre's credits in this panel

Producer	Creative authorship and production
Mixed by	Execution
Keyboards	Execution

Three tasks, one contributor

Dr. Dre works across two task categories on this record.

One contributor can perform several tasks



Credits

A&R [Director] – John McClain
Art Direction, Design – Kimberly Holt*, Unleashed (2)
Coordinator – Kimberly Brown (2)
Engineer [Assistant Mix] – Willie Will
Executive-Producer, Management – Suge Knight
Flute, Saxophone – Katisse Buckingham
Guitar – Chris Clairmont, Eric (The Drunk) Borders*
Keyboards – Justin Reinhardt
Keyboards, Bass Guitar, Keyboards [Bass] – Colin W
Mastered By – Bernie Grundman
Mixed By – Chris (The Glove) Taylor*
Mixed By, Engineer [Mix] – Greg (Gregski) Royal*
Photography By – Daniel Jordan (4)
Producer, Mixed By, Keyboards – Dr. Dre
Written-By [Uncredited] – Percy Carey

Credits for *The Chronic*, Discogs

Dr. Dre's credits in this panel

Producer	Creative authorship and production
Mixed by	Execution
Keyboards	Execution

Three tasks, one contributor

Dr. Dre works across two task categories on this record.

Credits identify attributed tasks, without measuring hours, earnings, or effort.

Back

One contributor can perform several tasks



The Chronic, original LP back cover (1992)

Credits identify attributed tasks, without measuring hours, earnings, or effort.

▸ Discogs credits

▸ Back

One contributor can perform several tasks



The Chronic, original LP back cover (1992)

Dr. Dre's printed credits

Produced By DR. DRE

Creative authorship and production

Credits identify attributed tasks, without measuring hours, earnings, or effort.

▸ Discogs credits

▸ Back

One contributor can perform several tasks



The Chronic, original LP back cover (1992)

Dr. Dre's printed credits

Produced By DR. DRE

Creative authorship and production

Mixed By: Dr. Dre,

Execution

Keyboards: Dr. Dre,

Execution

Credits identify attributed tasks, without measuring hours, earnings, or effort.

▸ Discogs credits

▸ Back

One contributor can perform several tasks



The Chronic, original LP back cover (1992)

Dr. Dre's printed credits

Produced By DR. DRE

Creative authorship and production

Mixed By: Dr. Dre,

Execution

Keyboards: Dr. Dre,

Execution

Three tasks, one contributor, working across two task categories.

Credits identify attributed tasks, without measuring hours, earnings, or effort.

▸ Discogs credits

▸ Back

Teams in recorded music: a Beatles example

Sgt. Pepper's Lonely Hearts Club Band (1967)

Creative authorship and production

Writing and composition





John Paul George

Production and arrangement



George Martin

Execution

Vocals and instruments






John Paul George Ringo

Additional instrumental performance



Orchestral and other session musicians

Recording engineering



Geoff Emerick

Nonproduction

Cover artwork



Peter Blake and Jann Haworth



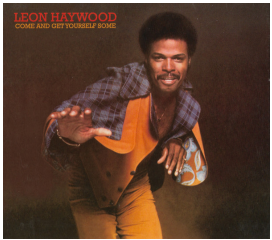
Peter Blake



Jann Haworth

Back

Recorded performances become inputs



Leon Haywood (1975)

I Want'a Do Something Freaky to You

▸ Play preview



Dr. Dre (1992)

Nuthin' but a "G" Thang

▸ Play preview

A recorded performance can become a musical input without being performed again.

▸ Back

What happened at the road clinics?



AKAI CLINICS were staged by three members of the Michael Jackson "Bad" tour. From left: keyboardist Greg Phillinganes, drummer Ricky Lawson, keyboardist Rory Kaplan.

From a drum pattern to a song

Greg Phillinganes, a keyboardist in Jackson's touring band, invited a child to program a drum pattern, then "turned it into a complete song."

What happened at the road clinics?



AKAI CLINICS were staged by three members of the Michael Jackson "Bad" tour. From left: keyboardist Greg Phillinganes, drummer Ricky Lawson, keyboardist Rory Kaplan.

From a drum pattern to a song

Greg Phillinganes, a keyboardist in Jackson's touring band, invited a child to program a drum pattern, then "turned it into a complete song."

Building a recording together

Ricky Lawson, the band's drummer, programmed drums and acted as recording engineer; Phillinganes recorded keyboard parts for a Stevie Wonder song.

Music Trades, October 1988, p. 39

▸ Back

Sample clearance can also generate authorship credits

December 1991: Grand Upright ruling

The ruling contributed to sample clearance becoming standard industry practice.

Sample clearance can also generate authorship credits

December 1991: Grand Upright ruling

The ruling contributed to sample clearance becoming standard industry practice.

Permission from the rights holders of sampled works

Writers of earlier works can receive credits without contributing to the new recording.

Sample clearance can also generate authorship credits

December 1991: Grand Upright ruling

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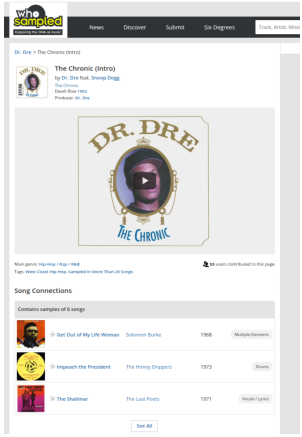
Writers of earlier works can receive credits without contributing to the new recording.

More authorship credits can reflect new contributions or additional attribution to earlier work.

Demers (2006); Grand Upright Music, Ltd. v. Warner Bros. Records Inc. (1991)

Back

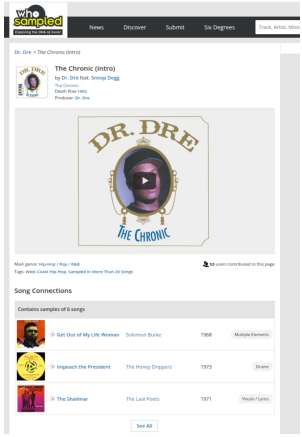
Sampling links: WhoSampled



Links to earlier recordings

The database identifies recordings and the earlier performances they sample.

Sampling links: WhoSampled



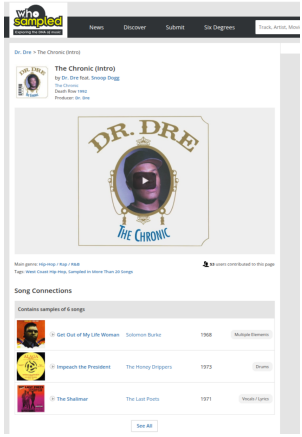
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The database identifies recordings and the earlier performances they sample.

Record-level sampling indicator

At least one track has a listed sample.

Sampling links: WhoSampled



Links to earlier recordings

The database identifies recordings and the earlier performances they sample.

Record-level sampling indicator

At least one track has a listed sample.

Retrospective coverage

A record without a listed sample may still use sampling.

Back

Linking contributors to backgrounds and recordings to content

Artist backgrounds: Wikidata and Wikipedia biographies

Musical training, education, birthplace, and birth year

Linking contributors to backgrounds and recordings to content

Artist backgrounds: Wikidata and Wikipedia biographies

Musical training, education, birthplace, and birth year

Recorded sound: Spotify Echo Nest and MusicFlamingo

Acoustic characteristics and audio embeddings

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Artist backgrounds: Wikidata and Wikipedia biographies

Musical training, education, birthplace, and birth year

Recorded sound: Spotify Echo Nest and MusicFlamingo

Acoustic characteristics and audio embeddings

Verbal content: Genius lyrics

Compare lyrical content across recordings.

Biographical characteristics remain unclassified when the source provides no information.

Back

Employment in music production

Distinct contributors in a commuting zone and year, per 100,000 residents aged 20–64

Measuring employment and the division of labor

Employment in music production

Distinct contributors in a commuting zone and year, per 100,000 residents aged 20–64

Team composition: people on each record

Creative authorship and production only, execution only, nonproduction only, or across categories

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Creative authorship and production only, execution only, nonproduction only, or across categories

Individual breadth

Working across task categories or holding multiple tasks within creative authorship and production

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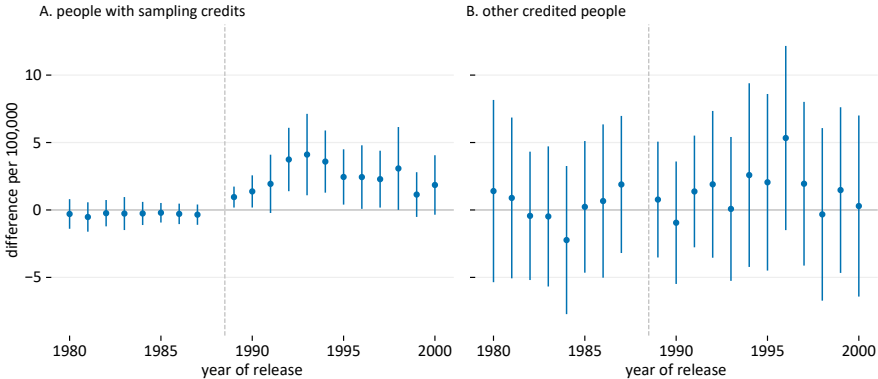
Working across task categories or holding multiple tasks within creative authorship and production

Task allocation

Share of credit lines in each of the three task categories

Back

Employment among contributors to sampled and other records



Distinct contributors per 100,000 residents aged 20–64; groups are not mutually exclusive; commuting-zone and division-by-year effects; reference: 1988.

Back

Team composition shifts toward creative authorship and production



Team composition shifts toward creative authorship and production

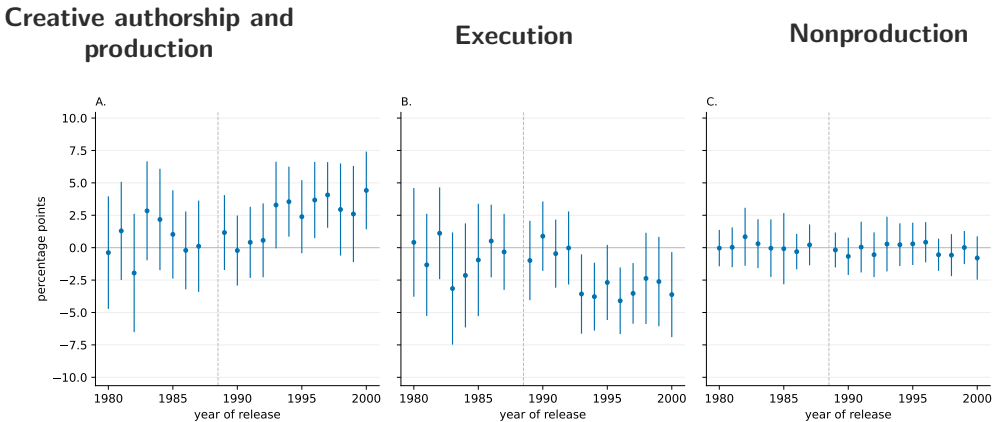


The creative-only share rises, without a detectable decline in execution-only contributor counts.

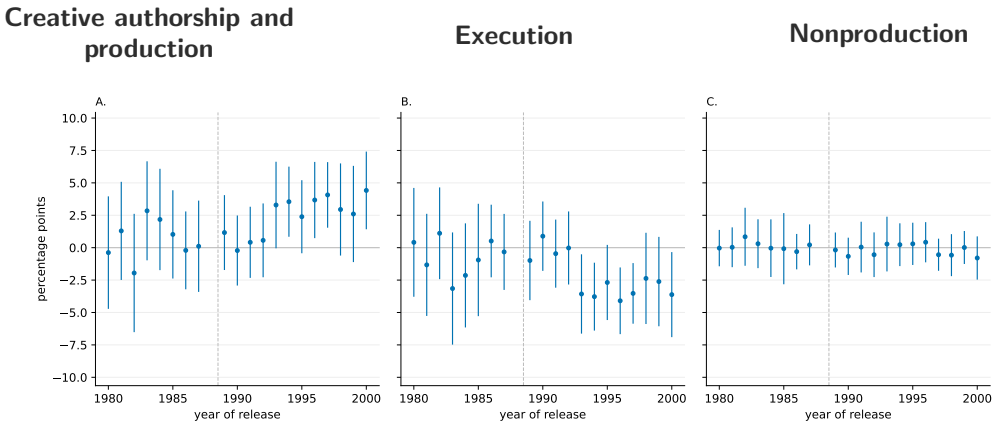
Studio and division-by-year effects; shares of all people on a record. Reference: 1988; 95% CR2 intervals; clusters: commuting zone.

Back

Task shares shift toward creative authorship and production



Task shares shift toward creative authorship and production

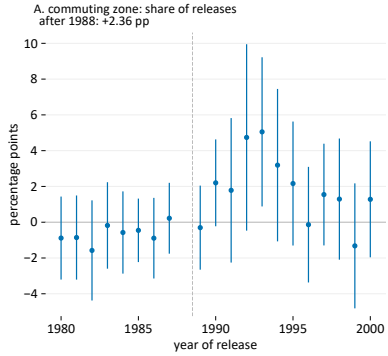


Creative task shares rise relative to execution, without a clear nonproduction response.

Studio and division-by-year effects; shares of credit lines, rather than people or hours. Reference: 1988; 95% CR2 intervals; clusters: commuting zone.

Back

Campaign exposure and the local share of records using sampling



The share of local records using sampling rises by **2.36 percentage points**.

One observation per active CZ-year. CZ and division-by-year fixed effects. Reference: 1988; 95% CR2 intervals; clusters: commuting zone.

Back

Diffusion: robustness checks

Local characteristics

Density, urban share, and income interacted with year preserve the early-1990s rise but reduce its gap from pre-campaign coefficients.

Diffusion: robustness checks

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Other tours and exposure reassignment

Other tours show no detectable response, while sampling reassignment p-values range from 0.006 to 0.021.

Diffusion: robustness checks

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Other tours and exposure reassignment

Other tours show no detectable response, while sampling reassignment p-values range from 0.006 to 0.021.

Local economic conditions

Employment, population, and income show no clear common expansion beginning with the campaign.

Population overlap is limited, with four exposed locations larger than every comparison location; reassignment is not randomization-based inference.

▸ Back

▸ Local controls ▸ Local economy ▸ Equipment listings

▸ Campaign-year coding ▸ Comparison-tour cutoffs

Employment: robustness checks

Local characteristics and functional form

Neither the linear rate nor Poisson counts with a population offset detect an employment response.

Employment: robustness checks

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Contributors and population separately

The contributor-count increase becomes imprecise with local controls, while population shows little differential movement.

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Population measurement

Checks compare annual with fixed-1980 denominators and remove fourteen imputed observations.

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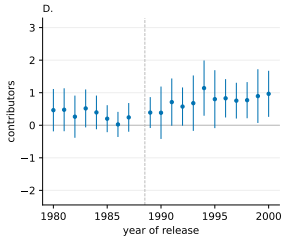
Collective entities

Removing entities always credited as groups changes annual estimates by at most 0.20 contributors per 100,000.

Back

Records gain contributors working only in creative authorship and production

Creative authorship and production only



Writing and composition account for most of the increase.

▸ Creative contributions ▸ Total team size

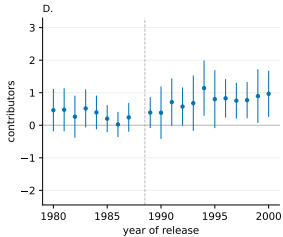
Studio and division-by-year effects; counts per record. Reference: 1988; 95% CR2 intervals; clusters: commuting zone.

▸ Robustness ▸ Sample attribution ▸ Sampling law ▸ Team shares ▸ Task shares

▸ Back

Records gain contributors working only in creative authorship and production

Creative authorship and production only



0.45 more creative-only contributors per record, about **13%** of the pre-campaign mean.

Writing and composition account for most of the increase.

▸ Creative contributions ▸ Total team size

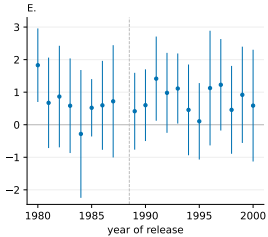
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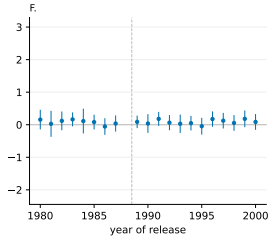
▸ Back

Records gain contributors working only in creative authorship and production

Execution only



Nonproduction only



No detectable offsetting decline in the other groups.

Writing and composition account for most of the increase.

▸ Creative contributions ▸ Total team size

Studio and division-by-year effects; counts per record. Reference: 1988; 95% CR2 intervals; clusters: commuting zone.

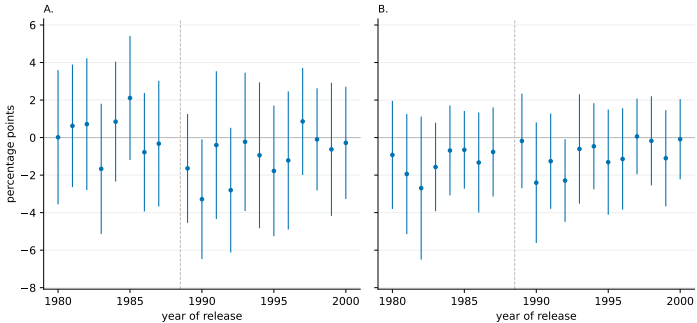
▸ Robustness ▸ Sample attribution ▸ Sampling law ▸ Team shares ▸ Task shares

▸ Back

No detectable broadening of individual task bundles

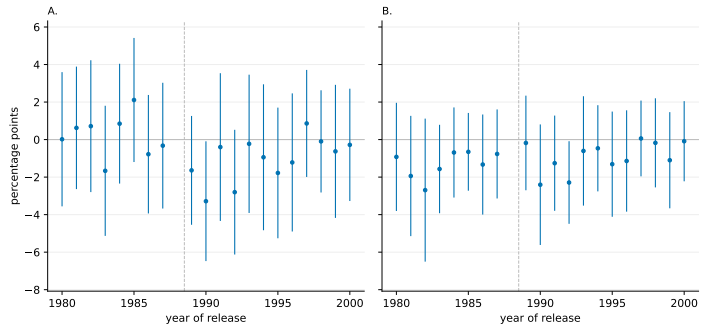
Working across task categories

Multiple creative tasks



No detectable broadening of individual task bundles

Working across task categories **Multiple creative tasks**



Integrating functions within a device does not translate into measurably broader individual task bundles.

Studio and division-by-year effects; shares of all contributors on a record. Reference: 1988; 95% CR2 intervals; clusters: commuting zone.

Back

The contributor increase survives alternative comparisons and assignments

Specification	Contributors	95% CR2 interval
Baseline	+0.449	[+0.081, +0.818]
Density, urban share, income × year	+0.382	[+0.130, +0.635]
Top third of pre-campaign production	+0.441	[+0.064, +0.817]
Top third of 1980 population	+0.426	[+0.036, +0.816]
Exclude records with earlier release dates	+0.554	[+0.189, +0.918]
All studios in one commuting zone	+0.249	[+0.022, +0.477]
Exclude ties across commuting zones	+0.293	[+0.066, +0.519]
Exclude all studio-assignment ties	+0.287	[+0.059, +0.516]
Missing credit categories assigned	+0.452	[+0.084, +0.820]
Reviewed credit-label assignments	+0.411	[+0.063, +0.759]

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The increase remains positive, although restricting records to one location reduces its magnitude.

Outcome: number working only in creative authorship and production; studio and division-by-year effects.

Back

How much could attribution to sampled works explain?

Remove writing and composition credits from sampled records released from 1992

	Contributors	95% CR2 interval
As observed	+0.449	[+0.081, +0.818]
Eligible credits removed	+0.352	[+0.032, +0.672]

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Removing these credits reduces the estimate by about a fifth.

The exercise also removes new writing, while unlisted samples can leave attribution credits in the data.

This is a sensitivity exercise, rather than a measure of how much attribution occurred.

Back

Songwriting comparison

Distance measures: construction

For recording i in genre g, released in year t:

$$D_i = 1 - \frac{1}{N_{t-1}} \sum_h w_{gh} \sum_{j \in h, t-1} \cos(v_i, v_j).$$

- v_i: normalized vector of eleven acoustic characteristics.
- w_{gh}: similarity between prior-year genre centroids, with $w_{gg} = 1$.
- N_{t-1}: number of measured recordings released in the previous year.

Record outcomes average measured track positions equally.

Lyrical distance uses the same comparison with vectors representing lyric content. MusicFlamingo audio embeddings provide an alternative acoustic representation.

Effects are reported in standard deviations across records released through 1987.

Back

No detectable change in commercial performance

Outcome	Estimate	95% CR2 interval
Entered the Billboard 200 (pp)	-2.18	$[-6.88, +2.52]$
Weeks on the Billboard 200	-0.41	$[-1.22, +0.41]$
Rank-weighted chart index ($\beta = 1$)	$+0.014$	$[-0.027, +0.054]$

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The estimates leave the magnitude of the commercial response uncertain.

Chart entry and performance measure relative market success, without measuring dollar revenue.

Back

Organization: additional robustness checks

Credit classifications and exclusions

Alternative label assignments and removing collective, executive-producer, or featuring credits retain a positive creative-contributor estimate.

▸ Sample attribution

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Timing, genre, and release format

Pre-period normalization shows the gradual adjustment, while genre effects change count estimates little overall and by format.

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▸ Sample attribution

Timing, genre, and release format

Pre-period normalization shows the gradual adjustment, while genre effects change count estimates little overall and by format.

Exposure reassignment

Creative-count reassignment p-values are 0.127, 0.132, and 0.034 across the three assignment schemes.

Neither alternative credit classification detects broader individual tasks; differences in contributor-count responses across formats are imprecise.

▸ Back

▸ Creative contributions ▸ Team size ▸ Task decomposition

Output: robustness checks

Fixed effects and comparison set

Studio effects give a similar acoustic estimate, while genre effects reduce it and within-genre comparisons show no detectable increase.

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MusicFlamingo embeddings give an imprecise estimate that does not clearly confirm the engineered-feature result.

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Lyric length and comparison window

Length correction and the earlier two-year comparison detect no lyrical response.

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Lyric length and comparison window

Length correction and the earlier two-year comparison detect no lyrical response.

Coverage

No detectable differential change in audio or lyrical coverage, with intervals that leave selection unresolved.

Individual-feature diagnostics use a different benchmark and sample, so they do not decompose the acoustic-distance response.

Back

Audio representation | Lyric length | Coverage

Audio coverage | Current lyric measures

Does hosting a major concert explain the sampling increase?

Exposure assignment	Estimate (pp)	95% CR2 interval
Sampler campaign	$+2.36$	$[+0.52, +4.19]$
Other major tours	$+0.62$	$[-1.16, +2.40]$

Does hosting a major concert explain the sampling increase?

Exposure assignment	Estimate (pp)	95% CR2 interval
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Initial local characteristics

Allowing density, urban share, and income to follow separate annual paths preserves the early-1990s rise, with smaller estimates.

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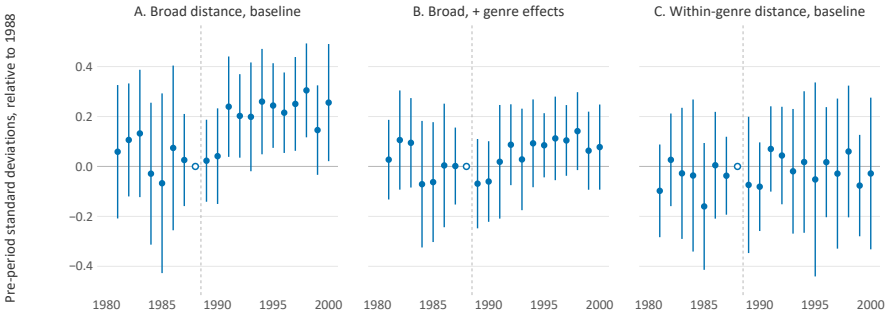
Broader local activity

No clear common expansion in employment, population, and income begins with the campaign.

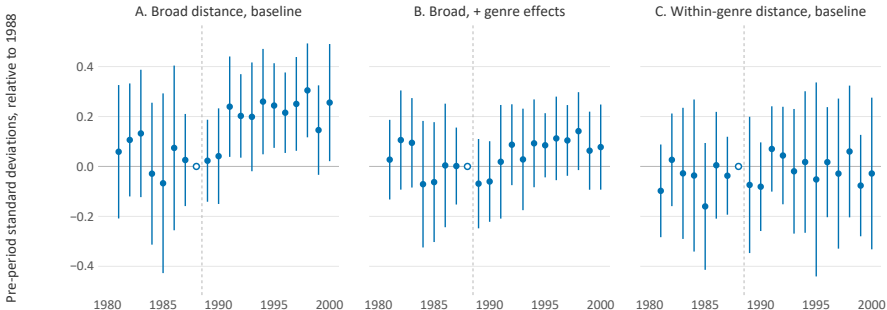
Other-tour assignments use the expanded comparison sample reported in the manuscript appendix.

Back

The acoustic shift concerns the broader set of recent releases



The acoustic shift concerns the broader set of recent releases



There is no detectable increase when the comparison is restricted to releases within the same genre.

Commuting-zone and division-by-year effects; adding primary-genre effects reduces broad distance to 0.05 SD. Reference: 1988; 95% CR2 intervals; clusters: commuting zone.

Back

From Discogs releases to the record sample

Restriction	Unit	Observations
Discogs releases in the extract	release	1,924,986
released 1980–2000	release	1,035,410
names a recording or mixing studio	release	208,126
main studio located to a US county	release	127,864
credit and sampling information present	release	124,300
Consolidated to records	record	77,077
primary genre inside recorded music	record	75,793
continental United States	record	75,594
commuting zone in the exposure design	record	70,999
at least one classified credit	record	70,994

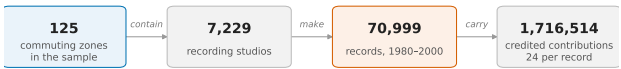
Releases sharing a master identifier are consolidated; releases without one remain separate records.

Back

Units of analysis and source coverage

A. Where a record sits

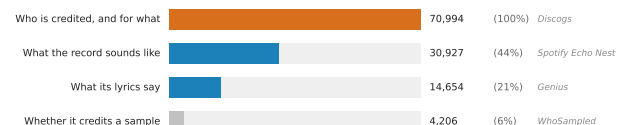
The campaign reached commuting zones; the outcomes are measured on the records made in them. All counts are the estimation sample.



18 of the 125 commuting zones were reached by the 1988 AKAI campaign; 8 comparison commuting zones host no record and enter only through commuting-zone outcomes.

B. What is observed about it

Share of the 70,999 records for which each source is available. One denominator throughout, so the bars are comparable; the samples the content measures are estimated on are smaller.



Commuting-zone population, income and employment are from the 1980 Census, BEA and SEER; the campaign's reach is from the 1988 tour itineraries.

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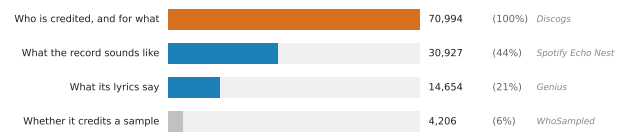
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Coverage differs across outcomes.

Source availability can exceed the number of records with a defined regression outcome.

Back

Record-level summary statistics before the campaign

Mean, 1980–1988	Exposed	Comparison
Records using sampling (%)	3.7	1.9
Creative-only contributors	3.402	3.311
Total contributors	15.9	14.0
Creative-only share (%)	25.6	27.5
Across task categories (%)	16.7	16.6
Multiple creative tasks (%)	10.7	9.8

Records weighted equally: 15,726 exposed and 8,418 comparison records with classified tasks.

Back

Exposed and comparison locations before the campaign

Mean	Exposed	Comparison
Population, 1980	3,792,387	779,670
Urban share, 1980 (%)	87.7	69.5
College share, 1980 (%)	19.1	16.1
Records per zone-year, 1980–1988	97.1	8.741
Active studios per zone-year	32.9	3.586
People per 100,000 aged 20–64	26.4	12.5

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Active studios per zone-year	32.9	3.586
People per 100,000 aged 20–64	26.4	12.5

Exposed locations were substantially larger recording centers.

18 exposed and 107 comparison zones; zone-year summaries weight zones equally, including inactive years where defined.

Back

Assigning records to studios and commuting zones

Recording and mixing studios

Use the studio with the most credit slots, excluding mastering locations and preferring recording over mixing when counts tie.

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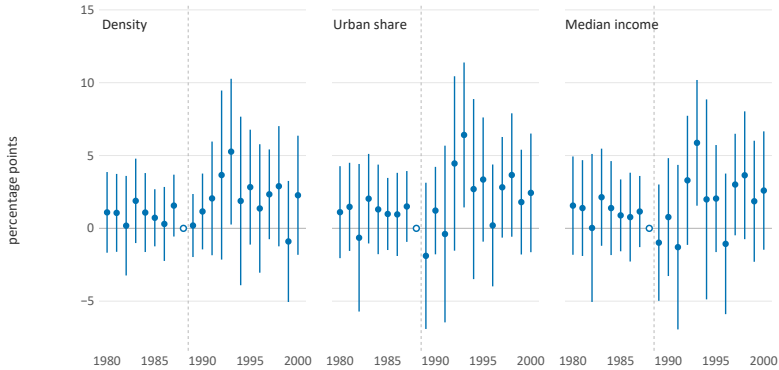
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Sensitivity to location assignment

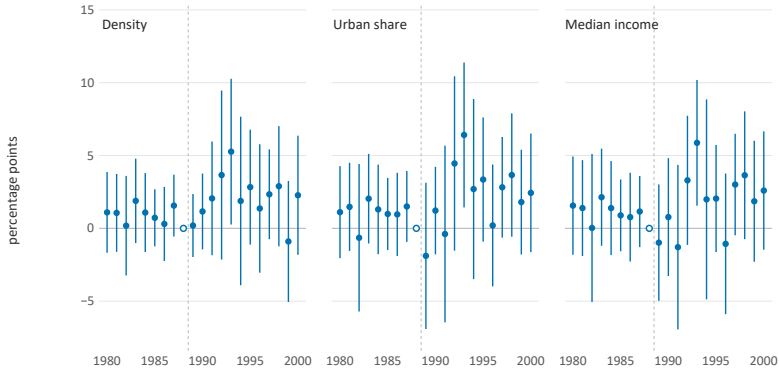
Restricting records to studios within one commuting zone retains 61,920 records and yields 0.25 additional creative-only contributors.

▸ Assignment checks ▸ Back

Sampling with predetermined local characteristics



Sampling with predetermined local characteristics

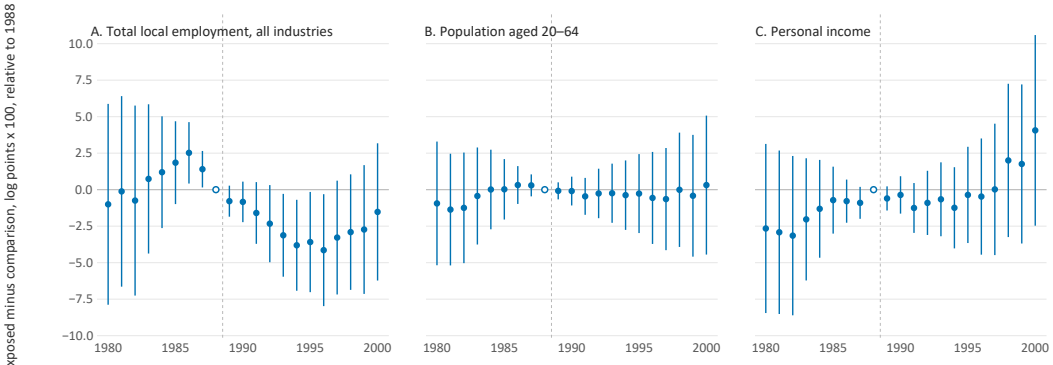


The early-1990s rise remains, but its gap from pre-campaign coefficients is smaller.

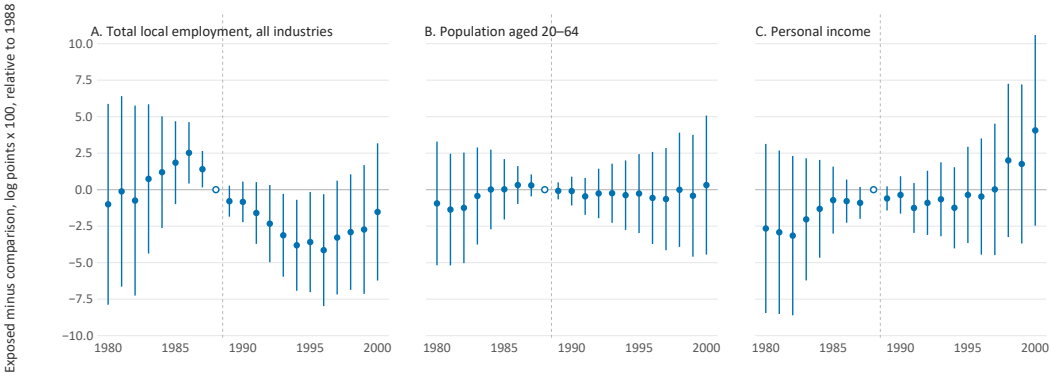
Each panel adds one predetermined characteristic interacted with year; CR2 intervals, clustered by commuting zone.

Back

Local employment, population, and income



Local employment, population, and income



No clear common local expansion begins with the campaign.

Annual patterns differ: employment moves before the campaign and income rises across both periods.

Back

Alternative treatment of the campaign year

Outcome	1988 pre	Drop 1988	1988 post
Creative-only contributors	$+0.45$	$+0.40$	$+0.30$
95% CR2 interval	$[+0.08, +0.82]$	$[+0.05, +0.74]$	$[+0.02, +0.59]$
Sampling (pp)	$+2.36$	$+2.42$	$+2.29$
95% CR2 interval	$[+0.52, +4.19]$	$[+0.51, +4.34]$	$[+0.46, +4.13]$

Alternative treatment of the campaign year

Outcome	1988 pre	Drop 1988	1988 post
Creative-only contributors	$+0.45$	$+0.40$	$+0.30$
95% CR2 interval	$[+0.08, +0.82]$	$[+0.05, +0.74]$	$[+0.02, +0.59]$
Sampling (pp)	$+2.36$	$+2.42$	$+2.29$
95% CR2 interval	$[+0.52, +4.19]$	$[+0.51, +4.34]$	$[+0.46, +4.13]$

Both increases remain under all three codings.

Release dates do not establish whether recording preceded the demonstration; baseline keeps 1988 in the pre-period.

Back

Number of tours defining comparison locations

Outcome	Ten tours	Eight tours	Five tours
Sampling (pp)	+2.357	+2.367	+2.347
People per 100,000	+3.85	+3.87	+3.69
Creative-only contributors	+0.449	+0.441	+0.441
Acoustic distance (SD)	+0.169	+0.167	+0.168
Comparison zones in design	107	106	103

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Comparison zones in design	107	106	103

Estimates are similar across the three cutoffs.

Cutoffs follow reported attendance order; they change the comparison group only slightly.

Back

What accounts for the creative-contributor increase?

Creative-only contributors	Estimate	95% CR2 interval
Writing and composition only	$+0.272$	$[+0.029, +0.516]$
Production and arrangement only	$+0.106$	$[-0.024, +0.236]$
Conducting only	-0.008	$[-0.020, +0.004]$
Across these creative components	$+0.079$	$[+0.027, +0.130]$
Total	$+0.449$	

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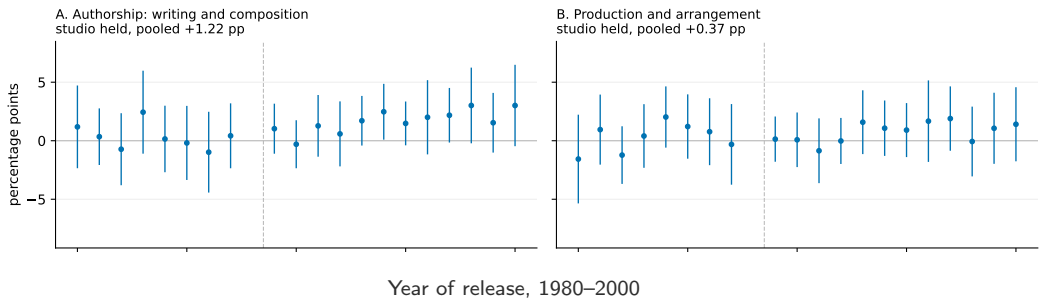
Writing and composition account for most of the increase.

People per record; studio and division-by-year effects; each contributor counted once.

Back

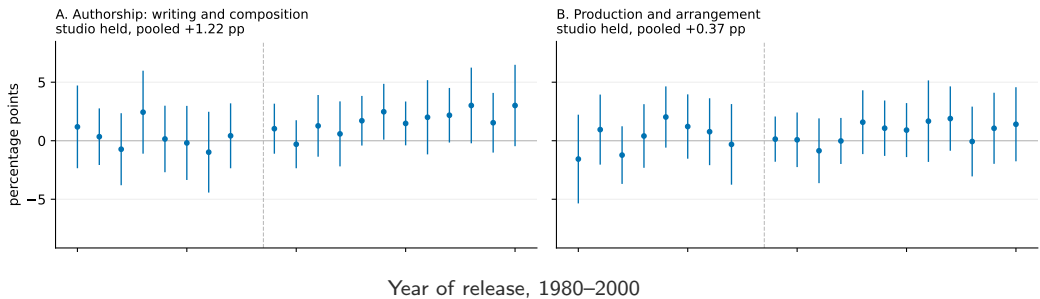
Writing and composition versus production and arrangement

The rising category split into its two blocks, by year



Writing and composition versus production and arrangement

The rising category split into its two blocks, by year



Authorship accounts for most of the shift in creative task shares.

Shares of credit lines; studio and division-by-year effects; 95% CR2 intervals.

Back

Contributor groups and total team size

Contributors per record	Pooled effect	CR2 SE	Pre-mean
Creative authorship and production only	+0.449	0.160	3.37
Execution only	+0.217	0.196	8.00
Nonproduction only	+0.023	0.071	1.62
Across task categories	−0.132	0.145	2.25
Total contributors	+0.556	0.406	15.24

Contributor groups and total team size

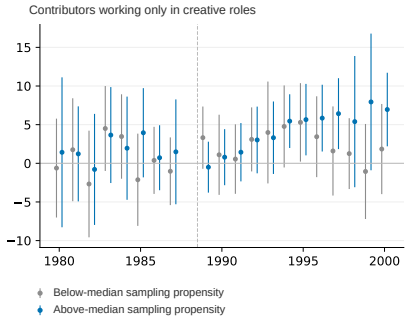
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The total-team-size estimate is imprecise.

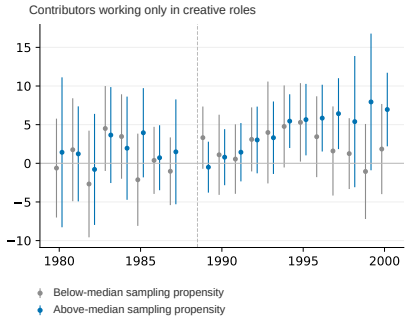
Studio and division-by-year effects; pooled 1989–2000 estimate; pre-means include 1988.

Back

Persistence in studios with greater prior sampling propensity



Persistence in studios with greater prior sampling propensity



The creative-only share remains elevated in higher-propensity studios.

Studio, division-by-year, and group-specific year effects; groups based on pre-campaign genres. Reference: 1988; 95% CR2 intervals; clusters: commuting zone.

Sampling may remain useful across successive projects, while earlier experience may help musicians use it and combine contributions.

▸ Specifications

▸ Back

Persistence: alternative specifications

Creative-only share (pp)	Estimate	95% CR2 interval
Baseline	$+5.0$	$[-3.0, +13.0]$
Clearance-eligible credits removed	$+3.6$	$[-4.3, +11.5]$
Genre-by-year effects	$+3.6$	$[-2.5, +9.8]$
Comparison-zone genre propensity	$+4.8$	$[-3.4, +12.9]$
Studio's own earlier sampling	$+7.7$	$[-6.6, +22.0]$
Drop one exposed zone: estimate range	$[+3.7, +6.0]$	

Persistence: alternative specifications

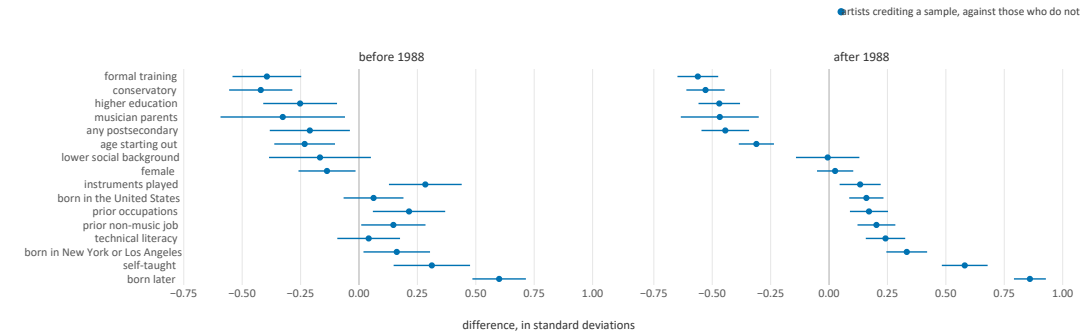
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The persistence pattern remains suggestive and imprecisely estimated.

Above-median studios: 1997–2000 average minus 1980–1987 average; shorter-window contrasts summarize the observed path.

Back

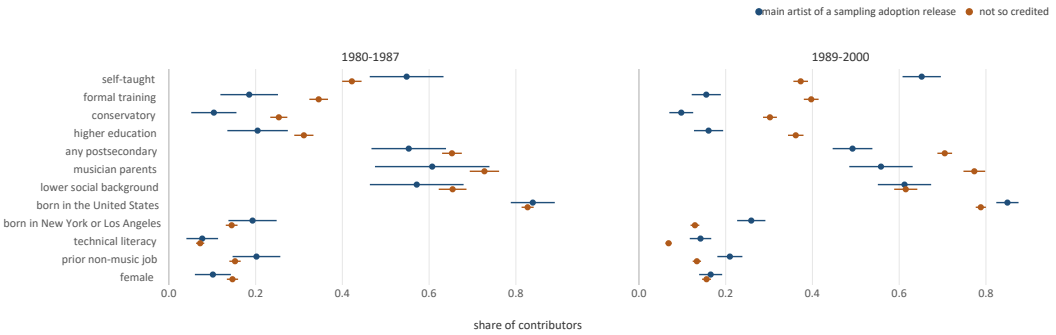
Artist backgrounds: full standardized comparisons



▸ Full group levels

▸ Back

Artist backgrounds: full group levels



Back

Training and coverage after 1988

Characteristic	Sampled artists	Other artists
Linked biographies	768	4,815
Training observed	451	3,175
Training coverage (%)	58.7	65.9
Self-taught among classified (%)	65.2	37.3
Formal training among classified (%)	15.5	39.7
Conservatory training (%)	9.8	30.2

Training and coverage after 1988

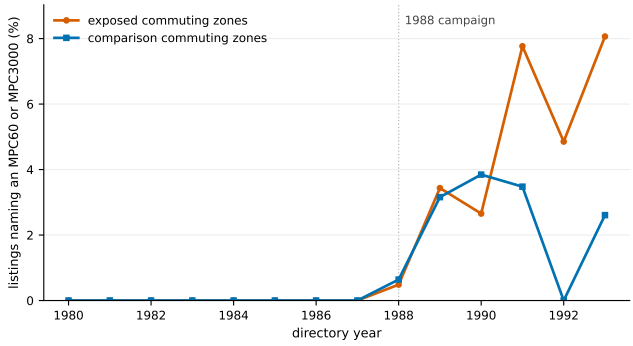
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The training comparison concerns artists with classified training information.

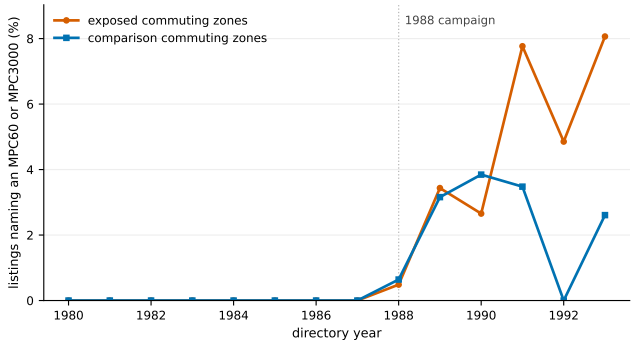
Each main artist contributes once per active period; sampling association means any sampled record in that period; comparisons are descriptive.

Back

MPC equipment listings in studio directories



MPC equipment listings in studio directories



Listing rates are similar through 1990, with a higher exposed share in 1991–1993.

Descriptive: 3,876 full-format listings name MPC60 or MPC3000; coverage and studios listed change over time.

▸ Back

▸ Campaign

Songwriting beyond attribution to reused material

Number-one songs	1980–84	1985–88	1989–91	1992–95	1996–2000
Number of songs	82	117	84	42	63
All songwriters	2.00	2.15	2.12	3.07	3.73
Excluding sampled/interpolated writers	2.00	2.12	2.04	2.52	3.08

Songwriting beyond attribution to reused material

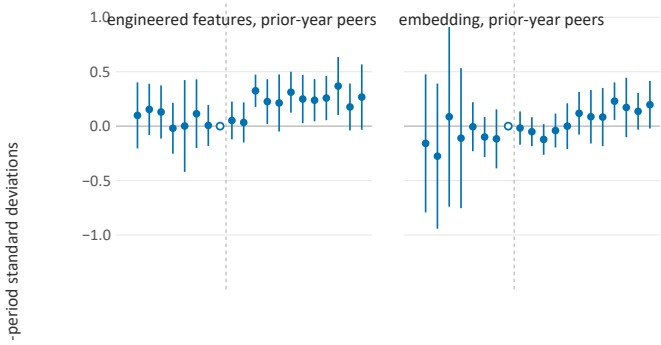
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Songwriting credits rise even after excluding writers attributed through samples or interpolations.

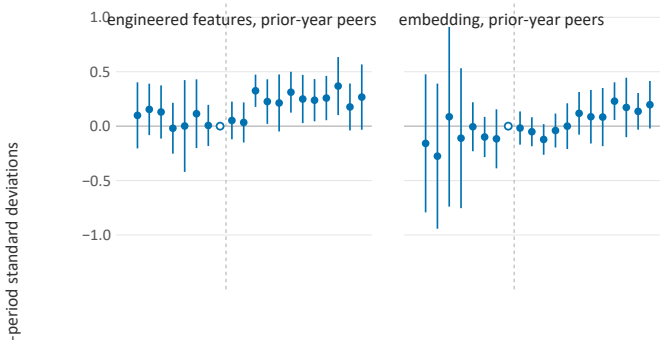
Dalla Riva (2025); means among Billboard Hot 100 number-one songs; descriptive evidence outside the campaign design.

Back

Alternative representations of recorded sound



Alternative representations of recorded sound

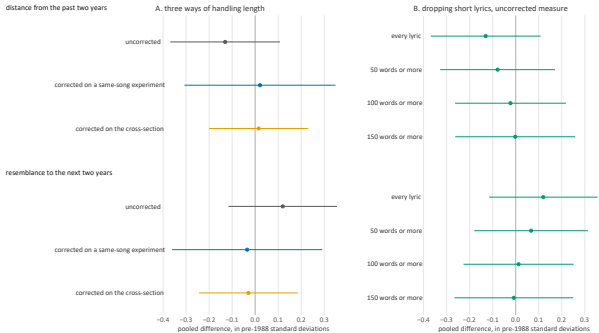


The embedding estimates are imprecise and do not clearly confirm the engineered-feature result.

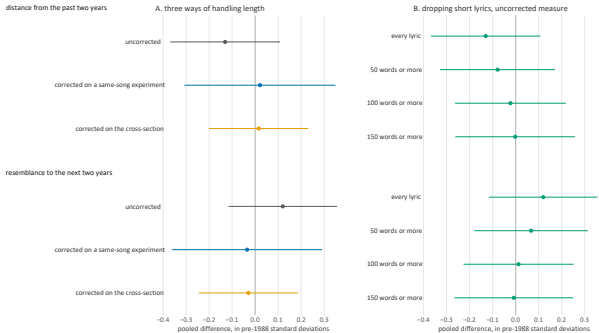
Common sample; MusicFlamingo audio embeddings; outcomes scaled by their pre-period dispersion.

Back

Sensitivity of lyrical distance to text length



Sensitivity of lyrical distance to text length

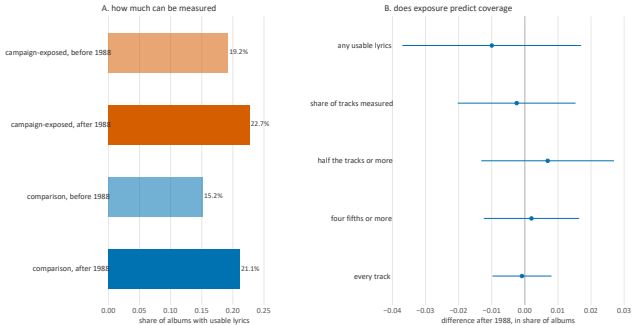


Length corrections and alternative comparisons detect no lyrical response.

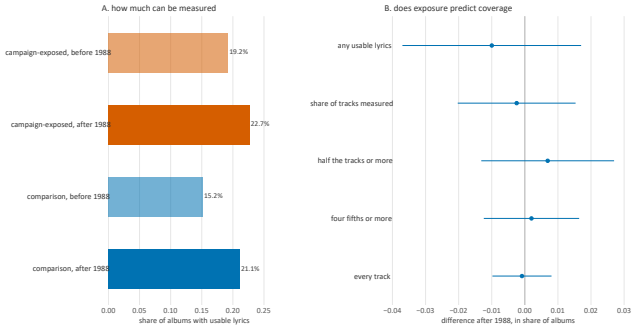
This diagnostic figure uses the earlier two-year comparison measures; the current outcome compares the previous year.

Back

Coverage of the lyrical-content measure



Coverage of the lyrical-content measure



No detectable differential change in lyrical coverage.

Coverage estimates remain imprecise and leave possible selection into the measured sample unresolved.

Back

Audio coverage following campaign exposure

Coverage outcome (pp)	Estimate	95% CR2 interval
Record has main acoustic features	-1.16	$[-5.79, +3.48]$
Share of tracks covered	-1.67	$[-6.49, +3.15]$

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Record has main acoustic features	-1.16	$[-5.79, +3.48]$
Share of tracks covered	-1.67	$[-6.49, +3.15]$

No detectable differential change in audio coverage.

67,510 records in 116 commuting zones; the estimates leave possible sample selection unresolved.

Back

Current lyrical-distance sensitivity

Lyrical distance (SD)	Estimate	95% CR2 interval
Uncorrected previous-year measure	$+0.023$	$[-0.337, +0.382]$
Length-corrected previous-year measure	$+0.195$	$[-0.273, +0.662]$
Length-corrected two-year measure	$+0.053$	$[-0.253, +0.359]$

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Length-corrected two-year measure	$+0.053$	$[-0.253, +0.359]$

No version detects a lyrical response.

Same 14,125 records; intervals condition on the estimated correction and exclude uncertainty from its estimation.

Back